

Sharp Extremal Gap Counts at Fixed Essential Period

A partial answer to Section 5.1 of arXiv:2112.15055

Autonomous proof packet for `fa_banach_001`

17 August 2026

Status. Candidate partial result, likely valid, pending human review. This packet proves a sharp extremal theorem for the source’s gap-versus-period question; it does not classify every intermediate gap pattern.

1 Source question

Kumar and Kumar ask for a relationship between the number of spectral gaps and the essential period of a self-adjoint periodic Jacobi operator and write that the number of gaps may be expected to be “on the order of” the period [1, Section 5.1, PDF p. 21].

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5.1. Number of gaps and the periodicity index. In the context of self-adjoint operators, it is imperative to conduct a detailed investigation into the relationship between the number of spectral gaps and the period. A notable example from [20] provides some insights into this connection, revealing that the number of spectral gaps in this case is precisely $p - 1$.

Consider the symbol $f_0(\theta)$ defined as follows:

$$f_0(\theta) = \begin{bmatrix} 0 & 1 & & & e^{-i\theta} \\ 1 & 0 & 1 & & \\ & 1 & \cdot & \cdot & \\ & & \cdot & \cdot & \cdot \\ & & & \cdot & 0 & 1 \\ e^{i\theta} & & & & 1 & 1 \end{bmatrix}.$$

In this specific example, the number of spectral gaps is established to be $p - 1$. Generally, one can expect that if A is a self-adjoint periodic Jacobi operator with an essential period of p , the number of gaps in the essential spectrum of A will be on the order of p .

For a two-sided self-adjoint Jacobi operator

$$(Ju)_n = a_{n-1}u_{n-1} + b_nu_n + a_nu_{n+1}, \quad a_n > 0, \quad b_n \in \mathbb{R},$$

the *essential period* is the least positive period of both coefficient sequences. An open spectral gap is a bounded component of the complement of the spectrum between two adjacent spectral bands.

2 Idea of the proof

The period is encoded not by the number of bands, but by their equilibrium (harmonic) masses. A two-band set can have one band of mass exactly $1/p$. Finite-gap inverse spectral theory then produces a periodic Jacobi matrix with that set as spectrum, and the reduced denominator p of the harmonic mass forces its least period to be exactly p . Thus one open gap survives at arbitrarily large periods. Börg's theorem rules out zero gaps when the least period exceeds one, so one is the sharp minimum. Standard Floquet theory gives at most $p - 1$ gaps and the source exhibits an example attaining that bound, giving the sharp maximum.

3 Finite-gap input

Let

$$\mathfrak{e} = [\alpha_1, \beta_1] \cup \cdots \cup [\alpha_{\ell+1}, \beta_{\ell+1}]$$

be a finite-gap compact set, and let $\rho_{\mathfrak{e}}$ be its logarithmic equilibrium measure. The quantities $\rho_{\mathfrak{e}}([\alpha_j, \beta_j])$ are its band harmonic measures.

We use the following standard consequence of the isospectral-torus construction of Christiansen, Simon, and Zinchenko [2, Corollaries 4.5 and 6.4].

Lemma 1 (finite-gap period criterion). *For every finite-gap set $\mathfrak{e}$ there are two-sided self-adjoint Jacobi matrices in its isospectral torus with spectrum $\mathfrak{e}$. The coefficient sequences of every member of the torus are periodic with period q if and only if all band harmonic measures belong to $q^{-1}\mathbb{Z}$. Consequently their least period is the least common denominator of those masses.*

The last sentence follows directly from the “if and only if” assertion: period q is possible precisely when multiplication by q makes every mass integral.

4 One gap at every least period

Lemma 2. *For every integer $p \geq 2$, there is a union $\mathfrak{e}_p$ of two disjoint compact intervals whose band equilibrium masses are $1/p$ and $(p-1)/p$.*

Proof. For $p = 2$, take any symmetric two-band set

$$[-1, -a] \cup [a, 1], \quad 0 < a < 1.$$

Reflection symmetry and uniqueness of the equilibrium measure give mass $1/2$ to each band.

Now let $p > 2$ and, for $0 < s < 1$, put

$$E_s = [-1, 0] \cup [s, 1], \quad h(s) = \rho_{E_s}([s, 1]).$$

The equilibrium measures vary weakly continuously when these interval endpoints vary. One quick justification is the standard energy-minimization argument: along a monotone sequence of endpoints, logarithmic capacity is continuous and every weak limit of the equilibrium measures has the minimal energy of the limiting compact set; uniqueness identifies that limit. An arbitrary convergent sequence has monotone subsequences, giving full continuity. Since the two bands remain separated near an interior parameter, their masses are continuous as well.

As $s \downarrow 0$, the compact sets tend to $[-1, 1]$ and

$$h(s) \longrightarrow \rho_{[-1, 1]}([0, 1]) = \frac{1}{2}$$

by reflection symmetry. As $s \uparrow 1$, the right interval collapses to the polar singleton $\{1\}$; the limiting equilibrium measure is that of $[-1, 0]$ and gives this singleton mass zero. Hence $h(s) \rightarrow 0$. The intermediate value theorem supplies $s_p \in (0, 1)$ with $h(s_p) = 1/p$. Taking $\mathfrak{e}_p = E_{s_p}$ proves the claim. $\square$

Theorem 1 (one-gap realization). *For every integer $p \geq 2$, there is a self-adjoint Jacobi operator of least period exactly p whose spectrum has exactly one open gap.*

Proof. Apply Lemma 1 to the two-band set $\mathfrak{e}_p$ from Lemma 2. Its band masses lie in $p^{-1}\mathbb{Z}$, so the isospectral torus gives a p -periodic Jacobi matrix J_p with $\sigma(J_p) = \mathfrak{e}_p$.

Its least period cannot be a positive integer $q < p$. Indeed, period q would force $1/p = k/q$ for an integer k , by Lemma 1; equivalently $q = kp$, impossible for $0 < q < p$. Thus the least period is p . Since $\mathfrak{e}_p$ consists of exactly two disjoint bands, it has exactly one open gap. $\square$

5 Sharp extremal theorem

For $p \geq 2$, let $g_{\min}(p)$ and $g_{\max}(p)$ denote respectively the minimum and maximum number of open gaps among self-adjoint Jacobi operators of least period p .

Theorem 2. *For every integer $p \geq 2$,*

$$g_{\min}(p) = 1, \quad g_{\max}(p) = p - 1.$$

Proof. Börg’s theorem for periodic Jacobi matrices says that all gaps are closed only when the coefficient sequences are constant [1]. A matrix of least period $p > 1$ therefore has at least one open gap. Theorem 1 attains one, proving $g_{\min}(p) = 1$.

Floquet theory represents the spectrum of a p -periodic Jacobi matrix as at most p compact bands, so there are at most $p - 1$ open gaps. The explicit matrix-valued symbol $f_0(\theta)$ in Section 5.1 of the source has least period p and exactly $p - 1$ gaps [1, p. 21]. Hence the upper bound is attained and $g_{\max}(p) = p - 1$. $\square$

Corollary 1. *No estimate of the form*

$$\#\{\text{open gaps of } J\} \geq F(p)$$

can hold for all self-adjoint Jacobi matrices of least period p if $F(p) > 1$ for any $p \geq 2$. In particular, no positive linear lower bound in p is possible.

6 Verification, scope, and novelty

The proof is analytic and uses no numerical computation. The decisive checks for a reviewer are (i) weak continuity and the endpoint limits of $h(s)$ in Lemma 2, and (ii) the exact-period inference from Corollaries 4.5 and 6.4 of [2]. The source PDF and the decisive supporting PDF are included locally.

The result determines the sharp minimum and maximum number of gaps at each least period. It does not classify all intermediate counts or band geometries, and it does not address the source’s separate non-normal pseudospectral inequalities.

A bounded run-index and arXiv/web search on 17 August 2026 used the exact and near-exact phrases “one gap p -periodic Jacobi”, “exactly one gap periodic Jacobi”, “minimal period one gap Jacobi matrix”, and “arbitrary period finite gap Jacobi”. No exact statement of Theorem 2 was found. The argument is an agent-identified synthesis of established finite-gap theory, so novelty confidence is moderate.

Human-review recommendation

Review is recommended. If the equilibrium-measure continuity lemma and the least-denominator reading of the finite-gap period criterion are confirmed, the packet is a complete sharp extremal answer to the quantitative gap-versus-period interpretation of the source question.

References

- [1] Krishna Kumar G. and V. B. Kiran Kumar, *Stability in Non-Normal Periodic Jacobi Operators: Advancing Börg's Theorem*, arXiv:2112.15055, revised 2024.
- [2] Jacob S. Christiansen, Barry Simon, and Maxim Zinchenko, *Finite Gap Jacobi Matrices, I. The Isospectral Torus*, arXiv:0810.3273, 2008.